
SKILLS

- **Python Data Engineering:** Python, SQL, Pandas, ETL/ELT, data ingestion, transformations, validation, reusable libraries
- **Databases:** PostgreSQL, MySQL, Snowflake, Redis, Elasticsearch, schema design, query construction, data modelling
- **Pipelines:** Kafka, Airflow, batch/stream processing, scheduling, backfills, data quality checks, lineage, replayability
- **Data Transfer:** AWS S3, HTTP APIs, REST APIs, file ingestion, audit logs, secure data handling, vendor feeds
- **Distributed Systems:** microservices, APIs, event-driven workflows, caching, parallel processing, resilient services
- **Cloud/Infra:** AWS, Kubernetes, Docker, Terraform, CI/CD, Linux, Git, GitHub Actions, production deployments
- **Reliability:** OpenTelemetry, ELK, CloudWatch, monitoring, alerting, SLOs, runbooks, production support
- **Collaboration:** researcher onboarding, requirements analysis, technical specs, ADRs, code reviews, stakeholder demos

EXPERIENCE

- **Royal Bank of Canada (RBC)** Toronto, Canada
Staff Software Developer & Enterprise Architect – Data / Platform Nov 2022 – Present
 - **High-Scale Data Pipeline Platform:** Architected telemetry platform using OpenTelemetry, Vector, Fluent Bit, Kafka, and Logstash ingesting 150 TB/day across 10,000+ systems; standardised schemas, transformations, routing, validation, alerting, and replayable ingestion patterns for downstream analytics.
 - **Reusable Frameworks & Libraries:** Built shared platform abstractions, onboarding guides, schema contracts, and reusable service patterns adopted by 20+ teams; reduced point-to-point integrations and improved quality, maintainability, and throughput for data producers and consumers.
 - **Researcher/Stakeholder Dataset Workflows:** Work directly with architecture, infrastructure, product, security, and risk stakeholders to analyze requirements, model data, define query-friendly structures, build technical specs, and iterate through demos and production feedback.
 - **Production Support & Reliability:** Own operational readiness for critical data components through CI/CD, monitoring, runbooks, incident triage, escalation paths, data-readiness checks, SLOs, and normal-hours production support for regulated environments.
 - **0-to-1 Platform Delivery:** Led production delivery of patent-pending blueprint platform (Node.js/TypeScript, React, REST APIs) with real-time data rendering, rules-based recommendations, auto-certification workflows, and secure service matching.
- **HealthCard (Acquired Startup)** Remote
Blockchain Architect & First Full Stack Engineer Aug 2021 – Jul 2022
 - **Data-Driven Product Build:** Joined as first engineer and built React/TypeScript frontend, Node.js/Express APIs, AWS EKS deployment, audit logging, verification workflows, and trust-score data models for real-time premium validation and fraud-resistant claims processing.
 - **Document Pipeline Integration:** Integrated transformer-based document classification and monitoring for medical/insurance records, reducing manual review by 60%; built production APIs, feedback loops, and HIPAA-aligned audit trails for credential verification.
- **Canada Revenue Agency (CRA)** Ottawa, Canada (Remote)
Software Developer Co-op Apr 2022 – Nov 2022
 - **Production Software Modernization:** Migrated federal tax platform UI from Apache Struts to React/TypeScript and contributed to backend APIs serving millions of Canadians with accessibility, quality gates, secure delivery, and maintainable code standards.
- **Dalhousie University** Halifax, Canada
Research & Teaching Assistant Jan 2021 – Aug 2022
 - **Algorithms, Systems & Cloud:** Researched call-stack decision algorithms and Python tracing tools for compiler/runtime analysis; TA for Algorithms, Software Development, and graduate Advanced Cloud Computing.

PROJECTS

- **Market Data Research Pipeline:** Python framework for ingesting market data, validating schemas, constructing query-friendly datasets, running backtests, and producing reproducible analytics for strategy research.
- **Portfolio Risk Optimization Framework:** Python research system for market data ingestion, portfolio allocation, Sharpe analysis, Monte Carlo simulation, risk-adjusted signals, and repeatable quantitative experiments.
- **Automatic Trading Algorithm:** Sub-second latency personal project integrating live brokerage APIs and real-time market data streams; momentum/mean-reversion signals with risk controls and performance monitoring.

EDUCATION

- **Dalhousie University** Halifax, Canada
MASc Applied Computer Science, GPA: 3.9/4.0 | TA: Algorithms, Advanced Cloud Computing Jan 2021 – Aug 2022
- **University of Toronto** Toronto, Canada
Enterprise Architecture Certification (3-semester program) Completed 2024
- **Gujarat Technological University** Ahmedabad, India
B.E. Computer Engineering, GPA: 3.92/4.0 Aug 2016 – Aug 2020